

Market Brain Grid

A multi-agent trading intelligence system for the Indonesia Stock Exchange and crypto markets — daily brief, deep analysis, actionable trading plans, and backtesting, orchestrated by a swarm of role-based Hermes bots. v1.1 adds Astra-grade output discipline and the MBG Lite onboarding tier.

Market Brain Grid · Multi-Agent Trading Intelligence

Hermes-orchestrated · IDX + Crypto · Laptop + Supabase

Table of Contents

1. Executive Summary and Product Vision	3
1.1 Vision statement	3
1.2 Goals and non-goals	3
1.3 Design philosophy	4
1.4 What is new in v1.1	4
2. Market Context and Problem Statement	5
2.1 IDX market structure (verified 2025-2026 rules)	5
2.2 Crypto regulatory shift	5
2.3 The broker API reality	6
3. Research Synthesis: Framework Analysis	7
3.1 TradingAgents — the evidence-backed topology	7
3.2 AutoHedge — the execution schema	7
3.3 Vibe-Trading — the rigor and extensibility model	8
3.4 MOSS skills + Hermes Agent — the runtime	8
3.5 Capability comparison and adoption decisions	8
4. System Architecture	9
4.1 Component responsibilities	10
4.2 Technology stack	11
4.3 Daily operating rhythm	11
5. Bot Roster and Agent Role Definitions	12
5.1 Analysis and decision bots	13
5.2 Infrastructure bots	13
5.3 Hermes mechanics mapping	14
6. Feature Specifications	14
6.1 F-01 Daily market brief	15
6.2 F-02 Deep analysis pipeline	15
6.3 F-03 Actionable trading plan cards	16
6.4 F-04 Backtesting engine	16
6.5 F-05 Paper trading portfolio	17
6.6 F-06 Telegram interaction layer	17
6.7 F-07 Web dashboard	17
6.8 F-08 Reflection memory and learning	18

6.9 F-09 Risk engine and kill-switch	18
6.10 F-10 Opportunity scanner (P3)	18
6.11 F-11 MBG Lite (onboarding tier)	19
7. Data Layer and Integrations	19
7.1 IDX market data plan	19
7.2 Crypto data plan	20
7.3 Postgres schema	20
8. Backtesting and Validation Framework	21
8.1 Cost and friction model	21
8.2 Metrics and benchmark	22
8.3 Validation suite	22
9. Risk Management and Market Rules	23
9.1 Position sizing	23
9.2 IDX microstructure guards	24
9.3 Compliance posture and disclaimer	24
10. Deployment and Infrastructure	24
10.1 Laptop worker (Docker Compose)	24
10.2 Secrets and key hygiene	25
11. Non-Functional Requirements and Cost Budget	25
11.1 Performance and reliability	25
11.2 Cost budget	26
12. Roadmap and Milestones	27
13. Success Metrics and Acceptance Criteria	28
13.1 Key performance indicators	28
13.2 Acceptance scenarios	29
14. Risks, Assumptions and References	30
14.1 Risk register	30
14.2 Assumptions	30
14.3 Open questions	31
14.4 References	31

1. Executive Summary and Product Vision

Market Brain Grid (MBG) is a personal multi-agent trading intelligence platform that covers the Indonesia Stock Exchange (IDX) and the crypto market from a single deployment. Every trading day it produces a structured morning brief, runs a deep multi-agent analysis on a user-defined watchlist, and publishes actionable trading plans with explicit entry, stop-loss, target, and position-size levels. A built-in backtesting engine replays historical decisions to measure whether the agent swarm actually adds value over simple benchmarks such as buy-and-hold on the IHSG composite index. The system is advisory by design in its first release: it simulates every recommendation in a paper-trading portfolio instead of touching real brokerage funds.

The product is engineered around the Hermes Agent runtime (Nous Research) as its orchestration layer. Each analytical role in the trading organization — technical analyst, fundamental analyst, sentiment analyst, news analyst, bull and bear researchers, trader, risk debaters, and portfolio manager — is modeled as a named Hermes bot profile with its own persona, model assignment, skills, and scheduled routines. This design choice is deliberate: Hermes bots are portable **SKILL.md** skill packages that already interoperate with the MOSS trade-bot ecosystem, OpenClaw, Claude Code, and Codex, which means the bot roster defined in this PRD remains usable even if the surrounding infrastructure is later replaced.

2% max risk

per trade — the hard position-sizing ceiling of the moderate risk profile

1.1 Vision statement

Give one Indonesian retail investor the research depth of an institutional trading desk: a team of specialized AI analysts that never sleeps, never gets bored, documents every argument it makes, and learns from its own past calls. MBG turns a lone laptop into a always-on research operation that scans the IDX and crypto markets before the opening bell at 09:00 WIB, watches for intraday technical triggers, and closes the day with an honest scorecard of what it recommended and what actually happened.

1.2 Goals and non-goals

The table below fixes the product boundary for version 1. Everything listed as a non-goal remains a possible future direction, but the architecture must not depend on any of them. This separation protects the critical path — a reliable daily analysis loop — from the temptation of premature automation.

In scope (v1 goals)	Out of scope (v1 non-goals)
Daily 07:00 WIB market brief covering IHSG, watchlist, and crypto	Real-money order execution on IDX brokers
Deep per-ticker multi-agent analysis with adversarial debate	High-frequency / sub-minute scalping strategies
Actionable trading plans with entry, stop, targets, position size	Multi-user SaaS tenancy and payments
Backtesting engine with walk-forward validation and benchmark alpha	Options, futures, and margin products on IDX
Paper-trading portfolio with IDX T+2 settlement simulation	Auto-execution of crypto orders (Phase 4+ feature, paper only in v1)
Telegram bot and web dashboard delivery channels	WhatsApp / SMS notification channels

Table 1 — Product goals vs non-goals for v1

1.3 Design philosophy

Three principles govern every requirement in this document. First, **advise before act**: the system earns trust through a measured paper-trading record before any real-money pathway is even designed. Second, **structure over prose**: every agent emits machine-readable structured outputs (JSON schemas for verdicts, plans, and risk opinions) so downstream code never has to parse free-flowing chat text; prose is reserved for the debate transcript and the human-facing brief. Third, **honest measurement**: every published recommendation is logged with its timestamp, the data available at decision time, and its eventual outcome, enabling the reflection memory and the backtester to compute genuine point-in-time alpha rather than flattering after-the-fact narratives.

1.4 What is new in v1.1

Version 1.1 absorbs the outcome of a comparative review of the ChatGPT Astra research guide (see the companion analysis document). Six output-discipline features were adopted into the core schemas because they close human-factor gaps at near-zero engineering cost, three larger items were scheduled onto the roadmap, and a new zero-infrastructure onboarding tier - MBG Lite - was added so the workflow delivers value before the full stack exists. The rebrand from the internal codename ATLAS to Market Brain Grid (MBG) applies to all product surfaces; bot profiles and skills use the mbg-* prefix throughout.

- **Adopted now**: FACTS vs OPINION separation in every report schema; the "NOT PROVIDED - verify manually" rule for missing data; a confidence rating (Low / Medium / High) with "what changes it" on every verdict and plan; a "numbers to verify" footer on plan cards; a "weakest assumption" field in every dossier; and approve / reject sign-off with notes on every plan.
- **Scheduled (P3)**: F-10 Opportunity Scanner (top-of-funnel idea generation from movers, sectors, and news) and the event-timing risk gate that blocks new entries into earnings or corporate-action windows.

- **Onboarding:** F-11 MBG Lite - a prompts-only Hermes skill bundle that runs the full workflow with manually pasted data from day zero, before the worker stack is deployed.

2. Market Context and Problem Statement

Indonesian retail investors face a structural information gap. The IDX lists more than 900 equities across the Main, Development, and Watchlist boards, yet quality research coverage concentrates on the 50-100 most liquid names. Sell-side reports are scarce, community discussion on Stockbit is noisy and unstructured, and global AI trading frameworks almost entirely ignore the market: as Chapter 3 documents, none of the four leading open-source trading agent projects ships an IDX data adapter. At the same time crypto trading in Indonesia has moved into a formal regulatory regime, creating both an opportunity and a compliance obligation that generic offshore bot templates do not address.

2.1 IDX market structure (verified 2025-2026 rules)

The operational rules below were verified against IDX publications and confirmed empirically by inspecting intraday bar timestamps from Yahoo Finance. They matter directly to bot scheduling, order-level price validation, and backtest realism, so the system treats them as configuration data rather than hard-coded assumptions.

Rule	Verified value	System impact
Session I	09:00-12:00 WIB (Fri 09:00-11:30)	Daily brief must land before 08:30 WIB
Session II (regular)	13:30-15:49:59 WIB (Fri 14:00-15:49:59)	Intraday trigger window for hybrid engine
Pre-closing / closing	15:50-15:59:59; closing match ~16:00-16:09:30	EOD snapshot scheduled 16:15 WIB
Tick fractions	Rp1 / Rp2 / Rp5 / Rp10 / Rp25 by price band	Limit-price rounding in plan cards
Auto rejection (ARA/ARB)	ARA 35% (=Rp5,000); ARB 15% (Main Board)	Order-price sanity check before user confirmation
Board lot	1 lot = 100 shares	Position size always expressed in lots
Settlement	T+2	Paper broker cash availability ledger
Illiquid stocks	Full call auction since Mar 2024	Liquidity filter excludes illiquid names from intraday engine

Table 2 — IDX trading rules adopted by MBG

2.2 Crypto regulatory shift

Oversight of crypto assets moved from Bappebti (Ministry of Trade) to OJK — with Bank Indonesia responsible for certain payment-related aspects — on 10 January 2025, with the full transition allowed up to 24 months under the P2SK law. OJK published its whitelist of 29 licensed trading platforms in

December 2025, including Indodax, Tokocrypto, Pintu, Reku, Triv, and Pluang. Binance global is not licensed to serve Indonesian residents, and regulators have actively blocked unlicensed platforms. MBG therefore treats the OJK whitelist as its execution universe: market data may be pulled from any source via the exchange-agnostic ccxt layer, but any future order routing (Phase 4+) will target whitelisted venues only. Trades on local exchanges also carry 0.1% final income tax plus 0.11% VAT under PMK 68/2022, which the backtester includes in its fee model.

2.3 The broker API reality

A defining constraint discovered during research: no Indonesian retail securities firm — Stockbit Sekuritas, Mirae Asset Sekuritas, IPOT, or others — currently publishes an official retail trading API for IDX equities. Unofficial scrapers exist but violate terms of service and put accounts at risk of suspension. This single fact reshapes the product: for IDX, MBG v1 is a decision-support system whose "execution" is a one-tap confirmation flow that hands a validated order ticket to the human user, while paper trading simulates the outcome. Crypto is the mirror image — both Indodax and Tokocrypto expose full trading APIs through ccxt (verified live during research) — but v1 still routes crypto through the paper broker to keep the trust-building sequence intact.

Pain today	MBG response
Scattered workflow: TradingView, Stockbit, CoinGecko, news sites checked manually every morning	Single 07:00 WIB Telegram brief consolidating index posture, watchlist signals, overnight crypto moves, and the day plan
No disciplined record of past calls, so the same mistakes repeat	Reflection memory logs every decision, scores it against IHSG alpha after 5 trading days, and injects lessons into future runs
Global AI trading tools have no IDX data adapters or local market rules	Purpose-built IDX data layer (yfinance .JK + idx.co.id scraper) with tick, lot, ARA/ARB, and T+2 logic
Backtesting LLM-driven ideas is slow and bias-prone	One-command signal-replay backtests with point-in-time memory filtering and walk-forward validation
Fear of handing funds to a black-box bot	Paper-first roadmap; nothing transacts without a human tap until Phase 4, and only then on OJK-whitelisted crypto venues

Table 3 — User pains mapped to MBG capabilities

3. Research Synthesis: Framework Analysis

Four open-source projects and one academic paper were analyzed in depth as the design baseline: TauricResearch/TradingAgents (plus its arXiv paper 2412.20138), The-Swarm-Corporation/AutoHedge, HKUDS/Vibe-Trading, and the MOSS trade-bot skill ecosystem together with the Hermes Agent runtime. The goal was not imitation but selection: identify the architecture patterns with published evidence behind them, the reliability tricks that prevent LLM-specific failures, and the gaps that MBG must fill for the Indonesian market.

3.1 TradingAgents — the evidence-backed topology

TradingAgents organizes an LLM trading firm as a 13-node LangGraph state machine: four analysts (market/technical, sentiment, news, fundamentals) feed a bull-versus-bear researcher debate; a research manager judges the debate on a five-tier scale; a trader converts the verdict into a transaction proposal with absolute price levels; a three-way risk debate (aggressive, neutral, conservative) stress-tests the proposal; and a portfolio manager issues the final decision. The paper reports the system outperforming MACD, KDJ+RSI, zero-mean-reversion, and SMA baselines on cumulative return, annualized return, Sharpe ratio, and max drawdown across AAPL, NVDA, MSFT, META, and GOOGL in early 2024. Version 0.4.0 of the codebase adds several production-grade ideas this PRD adopts wholesale: a pluggable data-vendor registry with ordered fallback chains, structured JSON outputs enforced only on decision nodes, a verified-market-snapshot tool that grounds any exact price claim (a direct anti-hallucination device), deterministic instrument-identity strings injected into prompts, checkpoint resume per ticker, and a two-layer reflection memory whose point-in-time filter prevents look-ahead bias during backtests. Its Apache-2.0 license permits full reuse.

3.2 AutoHedge — the execution schema

AutoHedge is a five-agent linear pipeline (director, sentiment, quant, risk manager, execution) built on the Swarms framework, notable for two ideas rather than its depth: ticker discovery from a natural-language task ("analyze the coal sector" yields a JSON array of tickers), and a compact order template — order type, quantity, entry price, stop loss, take profit, time in force — that maps cleanly onto an order-ticket data structure. It is also one of the few projects with real on-chain execution code (Solana via Jupiter Ultra), which informs the Phase 4 crypto execution design. Its weaknesses — single-pass agents with no debate, no memory, no backtesting, and hard-coded models — mark it as a source of schemas rather than architecture.

3.3 Vibe-Trading — the rigor and extensibility model

Vibe-Trading (HKUDS, roughly 33k GitHub stars, MIT) is a finance research workspace that turns natural language into data pulls, strategy code, backtests, and validation evidence. It ships 90 SKILL.md knowledge skills, 107 agent tools, a swarm DAG engine with 30 preset teams (including an investment_committee that mirrors the bull/bear/risk pattern), ten market-specific backtest engines with realistic microstructure (price limits, T+1/T+2 settlement, tick grids), Monte Carlo, bootstrap, and walk-forward validation, an alpha zoo of 462 prebuilt factors, and a hash-chained audit ledger. Two design cues are decisive for MBG: skills as editable markdown files with YAML frontmatter (the format Hermes and MOSS both consume), and the insistence that a missing model input makes a valuation not runnable instead of silently defaulted. Its most instructive gap: despite covering China, US, Hong Kong, India, Korea, Vietnam, and more, it has **no Indonesia support at all** — the gap MBG exists to close.

3.4 MOSS skills + Hermes Agent — the runtime

MOSS positions itself as an on-chain capital market for AI trading agents; its moss-trade-bot-skills repository (MIT-0) demonstrates the full SKILL.md package anatomy — frontmatter metadata, a deterministic markdown workflow, knowledge files, Python scripts, a skill.yaml manifest with declared permissions and config, and trigger keywords. Crucially, its README states compatibility with "OpenClaw, **Hermes**, Claude Code, Codex, and more." The Hermes Agent (Nous Research, MIT, Python 3.11+) is the strongest interpretation of the user's request to "run the bots with Hermes": it installs skills directly from GitHub with hermes skills install, turns profiles into named bots with personas (SOUL.md), model pins, and memory, provides cron routines and no-agent scheduled jobs, and orchestrates multi-bot work through group chats with @mentions, message_agent DMs, and cross-machine hermes peer. MBG therefore standardizes on Hermes as the bot runtime while keeping every skill package format-compliant with MOSS and OpenClaw.

3.5 Capability comparison and adoption decisions

Capability	TradingAgents	AutoHedge	Vibe-Trading	MOSS/Hermes	MBG
Multi-analyst fan-out	Yes (4 analysts)	Partial	Yes (30 teams)	Via skills	Adopted (4+1)
Adversarial debate	Yes (2 loops)	No	Yes (committee)	No	Adopted (2 loops)
Reflection memory	Yes (PIT-safe)	No	Partial	No	Adopted + alpha scoring
Backtest engines	None (paper-era loop)	None	Yes (10 markets)	Hyperliquid CSVs	vectorbt + IDX engine
Walk-forward / MC validation	No	No	Yes	No	Adopted

Capability	TradingAgents	AutoHedge	Vibe-Trading	MOSS/Hermes	MBG
SKILL.md packaging	No	No	Yes (90 skills)	Yes (native)	Adopted (native)
Structured outputs + grounding	Yes (v0.4)	No	Partial	No	Adopted
IDX market support	No	No	No	No	New — built here
Local exchange compliance	No	No	No	Hyperliquid only	OJK whitelist

Table 4 — Framework capability matrix

The adoption logic follows directly. From TradingAgents MBG takes the debate topology, the memory design, and the reliability layer, because they are the only patterns with peer-reviewed performance evidence. From AutoHedge it takes the order schema and the director-style task parsing for the Telegram /analyze command. From Vibe-Trading it takes the skill packaging discipline, the validation suite, and the backtest-metric conventions. From MOSS/Hermes it takes the runtime contract: bots are profiles, capabilities are skills, schedules are cron routines, and coordination happens over messages. Everything IDX-specific — data adapters, market rules, the benchmark map entry for ^JKSE, and broker-less execution flow — is original work that no existing project provides.

4. System Architecture

MBG uses a hybrid three-plane architecture that separates conversation, computation, and state. The **conversation plane** is the Hermes Agent gateway: a single long-running process that connects Telegram and the desktop app, hosts the bot profiles, installs MBG skills, and fires cron routines. The **computation plane** is the MBG worker — a Python 3.11+ application running under Docker Compose on the user's laptop — which hosts the LangGraph agent core, the data layer, the backtester with its paper broker, and the risk guard. The **state plane** is Supabase: a Postgres database as the single source of truth, Realtime channels that push updates to the dashboard, and object storage plus lightweight Edge Functions for scheduled end-of-day jobs. This split exists because serverless platforms cannot host a 24/7 Python agent (Supabase Edge Functions cap at 150-400 s wall-clock; Firebase gen-2 functions at 540-3,600 s), whereas a laptop worker with a restart policy and a database heartbeat handles it natively.

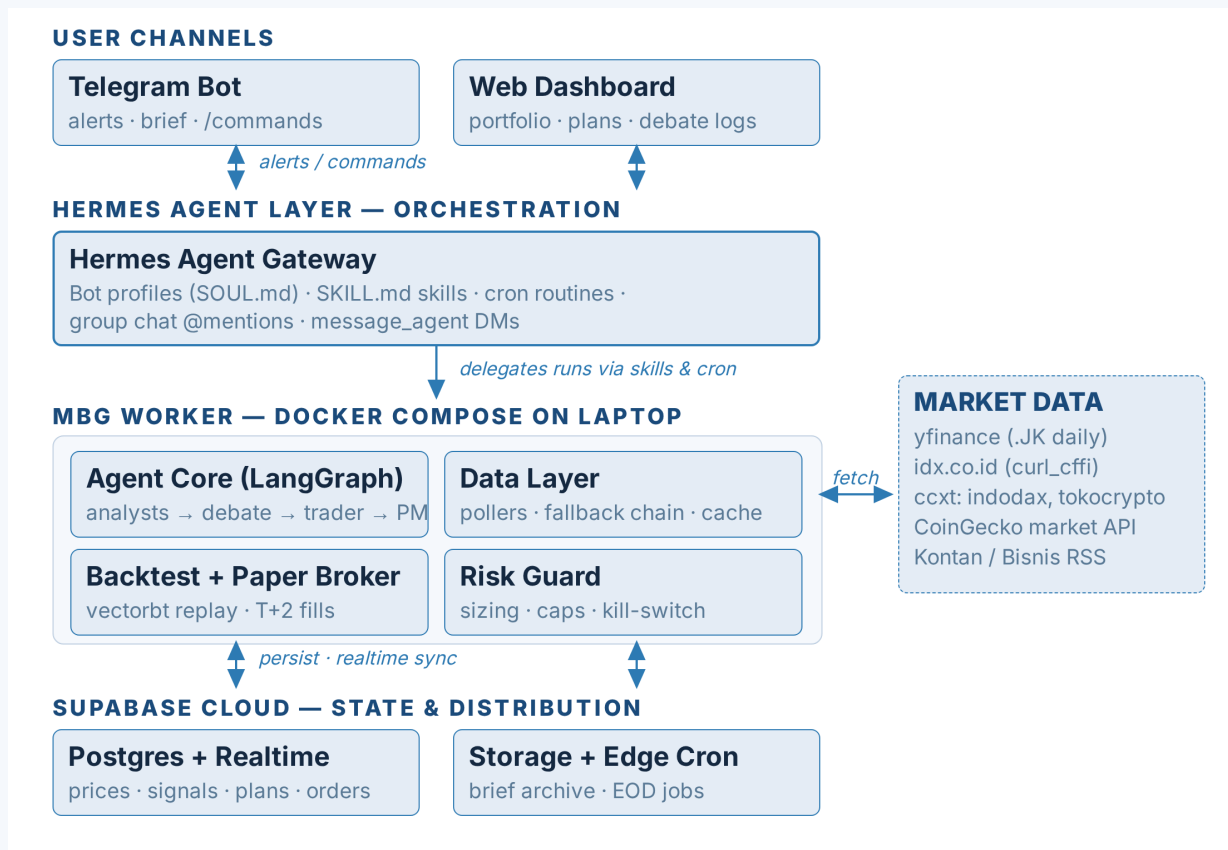


Figure 1 — MBG hybrid architecture: Hermes conversation plane, laptop worker, Supabase state plane

4.1 Component responsibilities

- **Hermes Gateway** — receives Telegram commands (/brief, /analyze BBKA, /plan, /status), routes them to the correct bot profile, delivers plan cards and alerts back to chat, and owns every cron routine (07:00 WIB brief, 16:15 WIB EOD, hourly crypto health check).
- **Agent Core (LangGraph)** — executes the 13-node analysis pipeline of Chapter 5 as a state machine with checkpoint resume per ticker; emits structured JSON at every decision node.
- **Data Layer** — scheduled pollers for each vendor, a SQLite/Parquet local cache, the ordered vendor fallback chain, and the point-in-time snapshot API that guarantees an agent only sees data available at decision time.
- **Backtest + Paper Broker** — vectorbt-based signal replay for LLM plans and deterministic strategies; simulated fills with IDX fees, IDR FX, slippage, and T+2 settlement; maintains the paper portfolio ledger.
- **Risk Guard** — deterministic pre-trade validation (Chapter 9 rules) that every plan must pass before publication; owns the kill-switch that halts all pipelines when drawdown or data-integrity limits trip.
- **Supabase** — Postgres schema of Chapter 7, row-level security, Realtime broadcast of plan and portfolio events to the dashboard, Storage bucket for brief archives, Edge cron for cloud-side EOD snapshots that run even when the laptop is briefly off.

4.2 Technology stack

Layer	Choice	Rationale
Bot runtime	Hermes Agent (Python 3.11-3.13, MIT)	Native SKILL.md skills, bot profiles, cron routines, Telegram gateway, multi-bot messaging
Agent graph	LangGraph >= 0.4	Proven 13-node topology, checkpoint/resume, conditional debate loops
LLM access	Provider-agnostic adapter (OpenAI / Anthropic / Google / DeepSeek / Ollama)	Per-role model pinning; vendor swap is a config change, not a code change
IDX data	yfinance (.JK) + curl_cffi scraper for idx.co.id	yfinance verified: BBKA.JK daily back to 2004; official site requires Cloudflare bypass
Crypto data	ccxt (indodax, tokocrypto) + CoinGecko free tier	Both exchanges live-verified via ccxt 4.5.78; CoinGecko demo = 10k credits/month
Backtesting	vectorbt 1.x (signals) + freqtrade (crypto strategy research)	Vectorized replay of LLM signal arrays; freqtrade actively maintained for crypto
State	Supabase Postgres 15 + Realtime + Storage	Relational fit for OHLCV and ledgers; generous free tier (500 MB, 50k MAU)
Dashboard	Next.js 14 + Tailwind (Vercel free tier)	Subscribes to Supabase Realtime; no server to operate
Packaging	Docker Compose (worker) + SKILL.md bundles (bots)	One-command laptop bring-up; skills portable to OpenClaw/MOSS/Claude Code

Table 5 — Stack selection and rationale

4.3 Daily operating rhythm

The hybrid engine runs two coordinated loops. The **daily loop** fires at 05:45 WIB with data refresh and integrity checks, 06:15 with per-ticker deep analysis of the watchlist (staggered to respect API rate limits), 07:00 with the consolidated Telegram brief, and 16:15 with the end-of-day snapshot that scores yesterday's plans, updates the paper portfolio, and queues memory reflections. The **intraday loop** polls five-minute bars for IDX during both sessions and continuous crypto feeds on a 15-minute cadence, evaluating only the lightweight trigger conditions attached to each active plan (breakout above entry zone, ATR-based stop breach, RSI extremes) — never the full LLM pipeline. When a trigger fires, the Risk Guard validates it and the Trader bot issues a plan-update card to Telegram within two minutes.

5. Bot Roster and Agent Role Definitions

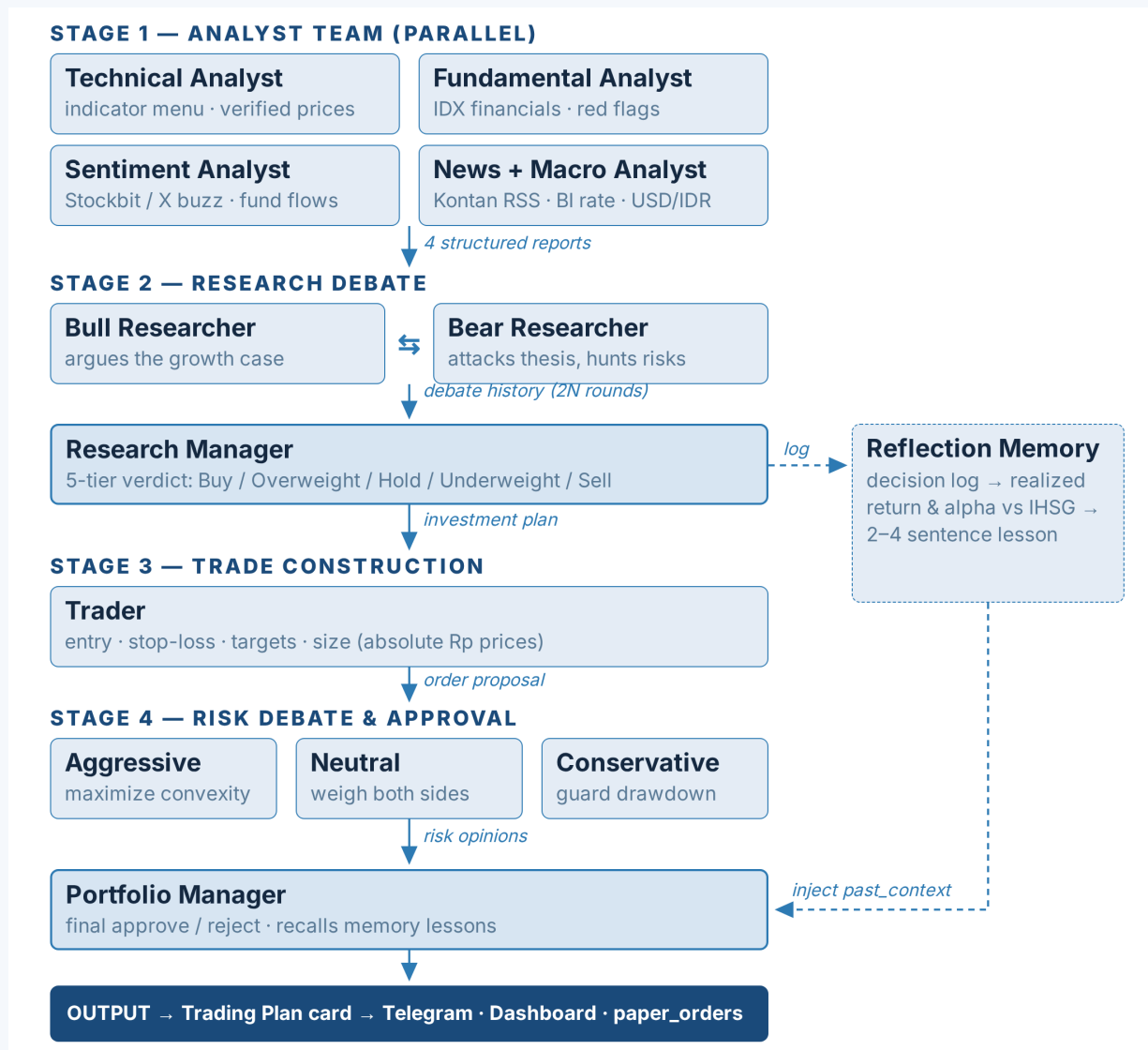


Figure 2 — The 13-node analysis pipeline with reflection memory loop (adapted from TradingAgents)

The analysis pipeline above is a direct adaptation of the TradingAgents topology, re-crewed for Indonesian markets and expressed as Hermes bots. Each bot is a Hermes profile: a named identity with a SOUL.md persona describing its mandate and tone, a pinned model, an allowlist of MBG skills, and its own memory namespace. Bots exchange structured reports through the shared LangGraph state rather than free-form chat, a communication protocol the source paper shows prevents the "telephone game" degradation of information as it passes between agents. Natural-language dialogue is reserved for exactly two places: the research debate and the risk debate, where adversarial rhetoric has genuine analytical value.

5.1 Analysis and decision bots

Bot (Hermes profile)	Model tier	Trigger	Inputs	Structured output
mbg-technical	fast	pipeline run	daily OHLCV, indicator menu, verified price snapshot	technical_report
mbg-fundamental	fast	pipeline run (IDX only)	IDX financial statements, ratios, earnings trend	fundamentals_report
mbg-sentiment	fast	pipeline run	Stockbit/X buzz proxies, funding and flow summary	sentiment_report (band, score, confidence)
mbg-news	fast	pipeline run	Kontan/Bisnis RSS, IDX disclosures, macro calendar	news_report
mbg-bull	fast	debate round	all analyst reports + debate history	bull_argument
mbg-bear	fast	debate round	all analyst reports + debate history	bear_argument
mbg-research-manager	deep	after N rounds	full debate transcript	investment_plan (5-tier verdict, rationale)
mbg-trader	fast	verdict received	investment_plan + technical_report	trader_proposal (action, entry, stop, targets)
mbg-risk-agg / -neu / -con	fast	risk debate	trader proposal + all reports	risk_opinion (approve/reject + reasons)
mbg-portfolio-manager	deep	risk debate closed	everything + memory past_context	portfolio_decision (final, publishable)

Table 6 — Analysis and decision bot roster

Model tiers translate to concrete models through a single configuration file: **fast** defaults to GPT-4o-mini or Gemini 2.5 Flash-Lite and powers all tool-using and debating roles, while **deep** defaults to Claude Sonnet 4.5 or GPT-4o and powers the two judges where reasoning quality compounds. The mapping is deliberately provider-agnostic: the entire roster runs unchanged on any provider in the adapter registry, including a local Ollama endpoint for zero-cost experimentation. Debate intensity is configurable — `max_debate_rounds` (default 1, meaning two exchanges) and `max_risk_discuss_rounds` (default 1, a three-way rotation) — so users can trade token cost against deliberation depth.

5.2 Infrastructure bots

Three non-analytical bots keep the operation honest and observable. **mbg-data-steward** owns the 05:45 WIB refresh: it walks the vendor fallback chain, validates bar counts and corporate-action adjustments, flags tickers whose data looks suspect, and publishes a data-quality row that the brief quotes verbatim — an

explicit design choice to make data problems visible instead of letting analysts reason over stale numbers. **mbg-risk-guard** is intentionally not an LLM bot in the approval path: it is a deterministic service (with a chat-facing profile for reporting) that enforces the Chapter 9 rulebook on every plan and every trigger, and holds the kill-switch. **mbg-reporter** assembles the 07:00 brief and end-of-day scorecard from structured outputs, renders the HTML archive to Supabase Storage, and never originates opinions of its own — it is a formatter, which keeps the brief faithful to what the swarm actually decided.

5.3 Hermes mechanics mapping

MBG concept	Hermes mechanism
Bot roster (15 profiles)	Hermes Bot Mode profiles under <code>~/hermes/profiles/</code> , each with SOUL.md persona, model pin, skill allowlist
Pipeline capability	mbg-pipeline SKILL.md skill that wraps the LangGraph runner CLI (also installable in OpenClaw/Claude Code)
Daily 07:00 brief, 16:15 EOD	hermes cron routines attached to the reporter profile (WIB timezone, no-agent mode invoking scripts directly)
Analyst-debate-trader handoffs	Group chat with @mentions for visibility; actual data handoff via shared state store, not chat text
Bot-to-bot requests (e.g. trader asks steward for fresh bars)	message_agent DMs with typed failure reasons
Cross-machine deployment (future VPS worker)	hermes peer bot-to-bot channel
Telegram commands	Gateway long-polling connection; skills register <code>/brief</code> , <code>/analyze</code> , <code>/plan</code> , <code>/status</code> , <code>/kill</code> commands

Table 7 — PRD concept to Hermes primitive

6. Feature Specifications

Nine features constitute the v1 product. Each is specified with its trigger, its inputs and outputs, and the acceptance criteria that the Phase exit reviews (Chapter 13) will verify. Priorities follow MoSCoW; the phase column anchors each feature to the roadmap of Chapter 12.

ID	Feature	Priority	Phase
F-01	Daily market brief (07:00 WIB)	Must	P1
F-02	Deep analysis pipeline (per ticker)	Must	P1
F-03	Actionable trading plan cards	Must	P2
F-04	Backtesting engine	Must	P2

ID	Feature	Priority	Phase
F-05	Paper trading portfolio (T+2 sim)	Must	P2
F-06	Telegram interaction layer	Must	P1
F-07	Web dashboard	Should	P3
F-08	Reflection memory and learning	Should	P3
F-09	Risk engine and kill-switch	Must	P2
F-10	Opportunity scanner (daily idea funnel)	Should	P3
F-11	MBG Lite (zero-infra onboarding tier)	Should	P0

Table 8 — Feature register

6.1 F-01 Daily market brief

The brief is the product's daily heartbeat and its primary retention surface. At 07:00 WIB the reporter bot publishes a single Telegram message (and dashboard banner) containing: IHSG posture with overnight global context; a one-line verdict per watchlist ticker from the previous evening's analysis; the top three actionable opportunities with plan-card links; crypto market summary (BTC and ETH in IDR via the reference exchange, top movers); today's economic calendar items affecting Indonesia (BI rate decisions, GDP releases, global events); and the data-quality line from the steward. The brief must be generated even when some agents fail — degraded sections are labeled as such rather than omitted, because a silently missing section reads as "nothing happening".

- **Inputs:** overnight data snapshot, latest per-ticker analyses, paper portfolio state, economic calendar feed.
- **Outputs:** Telegram message under 4,096 chars; HTML archive to Storage; brief row in Postgres.
- **Acceptance:** delivered by 07:05 WIB on $\geq 95\%$ of trading days over any 30-day window; every section present or explicitly marked degraded; archive URL resolves.

6.2 F-02 Deep analysis pipeline

One pipeline run produces a complete investment dossier for a single ticker: four analyst reports, the bull/bear transcript, the research manager's five-tier verdict (Buy / Overweight / Hold / Underweight / Sell), the trader's draft levels, the three-way risk debate, and the portfolio manager's final decision with an executive summary. Runs are checkpointed per ticker so a crashed LLM call resumes from the last completed node instead of restarting. The fundamentals analyst is skipped automatically for crypto assets (the TradingAgents asset-type filter pattern), and the news analyst always receives Indonesian-language sources for IDX tickers. A full run on 20 tickers must complete inside the 06:15-07:00 window on the default model configuration.

- **Inputs:** ticker, asset type, trade date, point-in-time data bundle, memory past_context.

- **Outputs:** dossier JSON (all node outputs), persisted to analyses table; optional Telegram dossier link.
- **v1.1 output rules:** every report separates FACTS (sourced, dated) from OPINION; any missing input is labeled "NOT PROVIDED - verify manually" instead of guessed; the dossier closes with a "weakest assumption" line naming the shakiest claim.
- **Acceptance:** p50 runtime ≤ 100 s per ticker on fast-tier models; resume works after injected node failure; every exact price claim in reports traceable to the verified snapshot tool.

6.3 F-03 Actionable trading plan cards

A plan card is the atomic unit of advice: ticker, action (Buy/Hold/Sell/Watch), entry zone (two absolute prices), stop-loss (absolute), two targets, position size in lots or USDT computed by the position-sizing formula of Chapter 9, risk-reward ratio, valid-until timestamp, and the plan's thesis in at most three sentences. Cards follow the AutoHedge-derived order schema with IDX-specific rounding to valid tick fractions and lot multiples. Every card carries a prominent **paper** badge in v1 and a one-tap "Simulate fill" action; IDX cards additionally show a pre-filled order ticket (prices, lots, estimated cost including fees) that the user can copy into their brokerage app. Trigger conditions attached to the card (for example "notify if price closes above 9,800" or "alert on 1x ATR adverse move") are evaluated by the intraday loop.

- **Inputs:** portfolio_decision + trader_proposal, portfolio state, risk-engine verdict.
- **Outputs:** plan row in Postgres, Telegram card with inline buttons, dashboard card event via Realtime.
- **v1.1 card fields:** confidence (Low / Medium / High) plus "what changes it"; a "numbers to verify before acting" footer; and Approve / Reject actions with an optional note, persisted on the plan row.
- **Acceptance:** 100% of published cards pass Risk Guard validation; prices snap to tick fractions; size respects 2% risk rule; expired cards auto-archive.

6.4 F-04 Backtesting engine

The backtester answers two different questions with two modes. **Signal replay** takes the historical stream of published plan cards (or any exported signal array) and replays entries and exits through vectorbt with IDX cost reality: 0.15-0.25% broker fee, 0.01% IDX levy, 0.1% PPh final on sell, 0.11% VAT on fees where applicable, plus configurable slippage and T+2 cash availability. **Strategy backtest** runs deterministic rule strategies (the same indicator logic the technical analyst uses) over ten years of .JK history or exchange OHLCV, so users can establish a strategy baseline before spending LLM tokens. Both modes enforce point-in-time integrity: memory injection filters on the as_of date, and the replay engine refuses any signal whose timestamp predates its data. Section 8 details the metrics and validation suite.

- **Inputs:** signal source (plans table or strategy spec), ticker universe, date range, cost model, benchmark (^JKSE or BTC/IDR).
- **Outputs:** run card with 15+ metrics, equity curve PNG, trade list, stored in backtest_runs.

- **Acceptance:** 10-year single-ticker daily replay completes < 60 s; costs match Chapter 8 tables to the fourth decimal; look-ahead sentinel tests pass.

6.5 F-05 Paper trading portfolio

The paper broker is the trust engine of v1: every published plan is tracked to outcome as if filled at the next session's opening auction, with realistic costs and IDX T+2 settlement (sell proceeds become available two trading days later; buying power respects the same rule). The ledger supports both IDX (lots, IDR) and crypto (fractional, USDT or IDR quote) positions under one equity curve. Daily mark-to-market at 16:15 WIB updates the portfolio, realizes stop/target hits with intrabar checks against five-minute bars where available, and feeds the scorecard section of the EOD message: hits, misses, open risk, and running alpha versus the benchmark. Users can reset the paper account or branch scenarios, but branches never merge back — preventing retroactive score flattering.

- **Inputs:** plan cards, five-minute and daily bars, cost model, calendar (holidays, sessions).
- **Outputs:** paper_orders and positions tables, equity curve, EOD scorecard.
- **Acceptance:** fill model matches declared rules on a golden test set of 50 scenarios; portfolio alpha recomputable from raw rows; T+2 availability honored (verified by unit tests simulating back-to-back trades).

6.6 F-06 Telegram interaction layer

Telegram is the primary control surface and uses Hermes's native gateway with long polling (the right default for a laptop behind NAT — no public endpoint or certificate is required). The command set: /brief (fetch latest or wait for the scheduled one), /analyze TICKER (queue a deep analysis, director-style natural language accepted, e.g. "/analyze bank stocks under 5000" expands via ticker discovery), /plan TICKER (latest plan card), /portfolio (positions, equity, open risk), /backtest TICKER RANGE (quick signal replay), /status (worker heartbeat, data freshness, model and cost meters), /watch ADD|REMOVE, /approve PLAN_ID and /reject PLAN_ID (sign-off with an optional note), and /kill (arm or release the kill-switch). Inline buttons on plan cards trigger simulate-fill, snooze, or dismiss. Broadcast etiquette follows the 30 messages/second API cap with a queue, which is inconsequential at single-user scale but keeps the design honest.

6.7 F-07 Web dashboard

The dashboard is a read-mostly Next.js application deployed on Vercel's free tier, authenticating the single owner via Supabase magic link and subscribing to Postgres Realtime. Views: portfolio overview (equity curve, drawdown, open positions with live-ish marks), plan cards board (active / triggered / expired / scored), the analyst debate transcript per analysis run (the transparency feature — users can read exactly why a verdict was reached), brief archive with rendered HTML, backtest run cards, and a system page (heartbeats, data freshness, token spend). The dashboard never writes trading state; its only mutations are

preferences and plan snooze/dismiss, keeping one write path (the worker) for integrity.

6.8 F-08 Reflection memory and learning

Memory follows the TradingAgents two-layer design with an MBG addition. Layer one is the **decision log**: after each pipeline run, the final decision is appended with its date, ticker, verdict, and a pending flag. Layer two is the **reflection**: five trading days later, the steward resolves the pending entry by fetching realized return and alpha versus ^JKSE (or the coin's benchmark), and prompts a cheap model to produce a 2-4 sentence lesson: was the directional call right, which part of the thesis held or failed, and one concrete instruction for next time. The MBG addition is **watchlist-level memory** — sector and macro lessons (for example "coal names ignore broad market rallies when China stimulus expectations fade") shared across tickers of the same sector. The portfolio manager receives the five most recent same-ticker lessons and three cross-ticker lessons as `past_context`, filtered to the `as_of` date during backtests. Memory is stored in Postgres (not local files) so it survives laptop reinstalls and is inspectable from the dashboard.

6.9 F-09 Risk engine and kill-switch

The risk engine is deterministic code, not an LLM, and sits in front of publication: a plan that violates any Chapter 9 rule is rejected with a machine-readable reason, and the rejection itself is logged and shown in the dashboard (rejected plans are visible — transparency includes seeing what the guardrails blocked). The kill-switch has three levels: soft (pause new analyses), hard (pause analyses and triggers), and panic (additionally flatten paper positions at next mark). It arms automatically on tripping conditions — portfolio drawdown beyond 10% from peak, three consecutive data-integrity failures, or daily loss beyond the 6% portfolio stop — and releases only by explicit /kill release with a typed confirmation phrase.

6.10 F-10 Opportunity scanner (P3)

The scanner is MBG's top-of-funnel, adopted from the Astra guide's SCAN step and industrialized. At 05:30 WIB, before the deep-analysis window opens, a lightweight non-LLM screen ranks the whole IDX universe (plus the crypto watchlist) on liquid heuristics: overnight movers with volume confirmation, foreign-flow outliers from the IDX trading summary, 52-week-high proximity, sector news velocity from the RSS corpus, and gap candidates. The top five candidates - each with a one-line reason and a link to its data - are appended to the morning brief, and any that the user promotes to the watchlist are queued for a full deep-analysis run. The scanner deliberately uses no LLM tokens in its ranking pass, keeping the daily cost ceiling intact; only promoted candidates consume pipeline budget.

- **Inputs:** prior-day trading summary, foreign flow, price/volume screens, sector news counts.
- **Outputs:** scanner section in the daily brief; promotion action writes to the watchlist.
- **Acceptance:** shortlist ready by 06:50 WIB; zero LLM tokens in ranking pass; every candidate traceable to its screening metric.

6.11 F-11 MBG Lite (onboarding tier)

MBG Lite is the zero-infrastructure expression of the product: a single Hermes skill bundle - installable with one command into Hermes, OpenClaw, or Claude Code - that contains the five-step workflow as SKILL.md prompts with the v1.1 output rules baked in. The user pastes dated data per ticker exactly as the Astra guide intends; the skill enforces the FACTS / OPINION split, the NOT PROVIDED rule, confidence labels, the verify-list, and the approve / reject ritual, and saves each finished plan as a markdown file in a local folder that later becomes the seed corpus for the full system's backtester. MBG Lite exists so the product is useful on day zero, and so the user builds the verification habit before automation removes the friction.

- **Contents:** mbg-lite SKILL.md (5-step workflow + master prompt), plan template, NEVER list.
- **Acceptance:** installs into Hermes with one skills command; produces a completed, rules-compliant plan for one manual ticker in under 15 minutes.

7. Data Layer and Integrations

The data layer's contract is simple: every consumer receives a consistent point-in-time DataFrame or JSON bundle regardless of which vendor served the request, and every fetch is cached, logged, and priced (tokens and API credits are metered per run). Vendor routing uses ordered fallback chains in the TradingAgents registry style — primary, secondary, tertiary — with automatic failover recorded in the data_quality log the steward publishes each morning.

7.1 IDX market data plan

Source	Coverage	Latency	Cost	Role
yfinance (.JK)	Daily/intraday OHLCV for IDX equities and ^JKSE, back to 2004 (verified)	~10-15 min delayed	Free	Primary EOD + intraday bars
idx.co.id (curl_cffi)	Trading summary, foreign flow, financial report PDFs, profiles	EOD	Free	Fundamentals, ownership, disclosures
Kontan / Bisnis RSS	Market news, editorials	Near-real-time	Free	News analyst corpus
IDX disclosures scrape	Corporate actions, halts, right issues	Event-driven	Free	Corporate action adjustments, halt calendar
Paid IDX API (optional)	Full market data via third-party vendors	Real-time	\$\$	Future upgrade if intraday edge needed

Table 9 — IDX data sources and roles

Two engineering cautions shape this plan. First, yfinance is an unofficial client of Yahoo endpoints: it is excellent for daily research bars (verified 5,486 rows for BBKA.JK back to June 2004 with Asia/Jakarta timezone) but subject to throttling, so the poller batches tickers, respects backoff, and never bursts. Second, idx.co.id sits behind a Cloudflare challenge that defeated plain HTTP clients during research; the steward uses a curl_cffi TLS-impersonation client with caching and treats HTML structure changes as first-class failure modes that trip the data-integrity alarm rather than silently returning empty fundamentals. Fundamentals update quarterly after laporan keuangan publication, so aggressive polling is unnecessary and discouraged.

7.2 Crypto data plan

The exchange-agnostic requirement lands on ccxt as the abstraction: indodax and tokocrypto (both OJK-whitelisted, both verified working through ccxt 4.5.78 during research — live BTC/IDR tickers pulled from each) serve as the default execution-grade venues, while any of ccxt's 100+ exchanges can be added as a data-only source by config. CoinGecko's free tier (10,000 credit credits/month, 100 requests/minute, verified against current pricing pages) covers global market caps, dominance, and trending data for the brief; CoinMarketCap's free 15,000 credits/month is the configured backup. On-chain and derivatives signals (funding rates, open interest) are Phase 4 additions that the pipeline's asset-type filter already anticipates.

7.3 Postgres schema

Table	Purpose	Key columns
instruments	Ticker/coin registry	symbol, venue, asset_type, sector, is_active, tick_fraction, lot_size
prices_daily	EOD OHLCV + adjustments	symbol, date, ohlcv, volume, adj_factor, source, fetched_at
prices_intra	5-minute bars (rolling 90d)	symbol, ts, ohlcv, source
fundamentals	Quarterly IDX financials	symbol, period, revenue, earnings, der, roe, eps, pbv, per, source_url
news_items	Deduplicated corpus	url, published_at, title, summary, source, tickers[], sentiment_hint
analyses	Pipeline dossiers	id, symbol, trade_date, node_outputs jsonb, model_config, tokens_in, tokens_out, cost_usd
plans	Plan cards	id, symbol, action, entry_lo, entry_hi, stop, targets[], size_lots, rr, thesis, valid_until, status, risk_verdict
paper_orders	Simulated fills	id, plan_id, side, qty, price, fees_idr, filled_at, settled_at (T+2)
positions	Open paper positions	symbol, qty, avg_cost, mark, unrealized, opened_from_plan
portfolio_daily	Mark-to-market equity	date, equity_idr, cash_idr, drawdown_pct, benchmark_close, alpha_pct

Table	Purpose	Key columns
memory_decisions	Decision log + lessons	symbol, date, verdict, pending, realized_return, alpha_return, lesson
backtest_runs	Run cards	id, mode, universe, range, metrics jsonb, equity_png_path, created_at
data_quality	Steward report rows	date, symbol, source_used, fallback_chain, bar_count, anomalies
heartbeats	Worker liveness	service, ts, version, kill_level

Table 10 — Core Supabase Postgres tables

Schema governance follows three rules. Every market-data row records its source and fetch time so any number in any report can be traced to its origin. Every LLM-generated row records its model configuration and token spend, making cost attribution per feature straightforward. And all timestamps are stored in UTC with the Asia/Jakarta offset applied only at presentation boundaries, eliminating the classic dual-timezone bugs that plague market software spanning WIB sessions and 24/7 crypto.

8. Backtesting and Validation Framework

Credibility is the product's scarcest resource, and the backtesting framework is where it is earned. The framework's design acknowledges an uncomfortable truth from the TradingAgents paper: LLM-driven backtests are not perfectly reproducible because model sampling and news corpora drift. MBG therefore separates what must be deterministic (the replay engine, cost model, and metrics) from what is inherently probabilistic (the agent verdicts), and reports confidence accordingly. Signal replay uses vectorbt for its vectorized speed — a 10-year, 20-ticker daily replay finishes in seconds — while freqtrade remains the recommended workbench for users who want deep crypto strategy research with hyperopt; NautilusTrader is the noted upgrade path if event-driven fidelity (order-book-level fills) ever becomes a requirement, and backtrader is explicitly avoided as effectively unmaintained since 2023.

8.1 Cost and friction model

Market	Cost component	Value	Notes
IDX buy	Broker fee	0.15-0.25% (config)	Retail online rate band
IDX buy	IDX levy + VAT effect	~0.01% + 11% on fee	Rounded to 0.011% effective on fee
IDX sell	PPh final	0.1%	Withholding on sale value
IDX both	Slippage (default)	0.1%	User-tunable; higher for illiquid boards
IDX both	Settlement	T+2 cash availability	Affects buying power, not P&L;
Crypto (local)	Exchange fee	0.3-0.5% taker (config)	Indodax/Tokocrypto retail bands
Crypto (local)	Tax + VAT on trade	0.1% PPh final + 0.11% VAT	PMK 68/2022, local venues

Table 11 — Transaction cost model used in replays

8.2 Metrics and benchmark

Every run card reports the standard quartet from the source paper — cumulative return, annualized return, Sharpe ratio, maximum drawdown — plus Sortino, win rate, profit factor, average win/loss, exposure time, turnover, and per-contribution attribution. The mandatory benchmark is ^JKSE for IDX universes and the reference exchange's BTC/IDR for crypto; alpha and beta against the benchmark are computed with the same risk-free assumption (BI rate, configurable) across all runs so numbers are comparable. A run without its benchmark row is considered invalid and the UI marks it as such — a deliberate friction against cherry-picking.

8.3 Validation suite

Three validation tools, adopted from the Vibe-Trading rigor model, are mandatory before any strategy or configuration earns a place in the daily loop. **Walk-forward analysis** splits history into rolling train/test windows (default: 12-month train, 3-month test) and reports the distribution of out-of-sample alpha, exposing regimes where the approach works and where it bleeds. **Monte Carlo perturbation** reshuffles trade order and applies return bootstrapping to put confidence bands on max drawdown — the honest answer to "was that 8% drawdown luck or design". **Look-ahead sentinels** are automated canary tests: the suite plants future-dated rows and memory entries and asserts the engine ignores them, directly guarding the point-in-time contract that makes paper-trading comparisons meaningful.

9. Risk Management and Market Rules

The risk rulebook encodes the moderate profile selected for this product: meaningful participation in upside with hard ceilings on single-decision damage. The numbers below are defaults stored in `config/risk.yaml`, editable in one place, and every deviation is logged. The first four rules are absolute in v1 — the Risk Guard rejects any plan violating them regardless of how confident the swarm is.

Rule	Default	Enforcement
Max risk per trade	2% of equity	Position size formula cap; rejects oversized plans
Max single-name exposure	20% of equity	Blocks entries that breach the cap incl. existing position
Max sector exposure (IDX)	35% of equity	Blocks entries; sector map from instruments registry
Daily portfolio stop	-6% from prior close	Arms soft kill; new longs blocked until next session
Max drawdown kill	-10% from equity peak	Arms hard kill until manual release
Mandatory stop-loss	Every plan	Cards without stops are invalid by schema
Min RR ratio (swing)	1.5	Rejects plans whose targets justify less
Max open plans	8 concurrent	Queues additional signals as Watch instead
Crypto leverage	Spot only in v1	Margin/perps out of scope until Phase 4 review
Event-timing gate (P3)	No new entries into earnings / corporate-action windows	Risk Guard blocks with reason; calendar from IDX disclosures scrape

Table 12 — Risk rulebook (moderate profile)

9.1 Position sizing

Position size follows the fixed-fractional formula $\text{size} = (\text{equity} \times \text{risk_pct}) / (\text{entry} - \text{stop})$, floored to whole IDX lots or crypto minimums, and then clamped by the exposure caps. A worked example anchors the convention: equity Rp 100,000,000; a BBKA plan with entry Rp 9,800 and stop Rp 9,500 risks Rp 3,000 per share; the 2% rule budgets Rp 2,000,000 of risk; the formula yields 666 shares, floored to 6 lots (600 shares, Rp 5,880,000 notional, ~5.9% of equity, actual risk Rp 1,800,000 or 1.8%). The plan card shows both the formula result and the floored execution size so the user always sees what the math wanted and what the lot grid allowed — a small transparency detail that teaches good sizing habits.

9.2 IDX microstructure guards

Order tickets are validated against live market rules before presentation: limit prices snap to valid tick fractions (Rp 1 below 200, Rp 2 to 500, Rp 5 to 2,000, Rp 10 to 5,000, Rp 25 above), and the ticket warns when an entry or stop sits within the day's auto-rejection bands (ARA 35% / 25% / 20% by price tier, ARB 15% on the Main Board) where it would be rejected by the exchange. Friday's compressed schedule (Session I ends 11:30, Session II starts 14:00) and the pre-closing/closing auction window are first-class calendar facts: triggers never fire in the final ten minutes of regular trading, and EOD jobs wait for the 16:15 close file. Illiquid names under the call-auction regime are excluded from the intraday engine entirely.

9.3 Compliance posture and disclaimer

MBG v1 is a research and education tool. It holds no client funds, transmits no orders, and publishes simulated performance only. IDX interaction ends at a copied order ticket; crypto auto-execution, when it arrives in Phase 4, will be opt-in, limited to OJK-whitelisted venues, capped by the same risk rulebook, and default to the smallest viable size. All published outputs carry a standing disclaimer — educational content, not financial advice; LLM outputs can be wrong; the user bears full responsibility for any action taken. This posture keeps the operator inside personal-use norms while the paper-trading record accumulates.

10. Deployment and Infrastructure

The deployment topology is laptop-first with a cloud state plane, and it is the direct consequence of one verified fact: neither Supabase Edge Functions (150-400 s wall-clock) nor Firebase Cloud Functions (540-3,600 s) can host a long-running Python agent process. Supabase wins the state-plane role over Firebase for three concrete reasons: a relational Postgres fits OHLCV and ledger data far better than Firestore documents; the free tier is more generous for this shape of load (500 MB database, 50k monthly active users, 5 GB egress, unlimited API requests versus Firestore's 50k reads/day cap); and Postgres Realtime provides a clean subscription channel for the dashboard. Firebase remains a documented alternative for users already invested in the Google ecosystem, supported by the same worker code writing to Firestore via an adapter — the worker talks to a storage interface, not to Supabase directly.

10.1 Laptop worker (Docker Compose)

The worker ships as a single docker-compose.yml with four services: mbg-worker (the Python application), hermes (the agent gateway), redis (debounce and queue for Telegram delivery), and caddy or nothing at all — no inbound ports are required because Telegram uses long polling and Supabase is outbound-only. Restart policy is unless-stopped on every service; the worker writes a heartbeat row every 60 seconds and a watchdog inside the worker restarts a pipeline node that has been silent for more than its checkpoint timeout. Volumes persist the local Parquet cache and Hermes profile state. The compose file is the entire laptop footprint: a fresh machine goes from git clone to first brief in under 15 minutes with only Docker, a

Supabase project, and API keys as prerequisites.

Aspect	Configuration
Worker process	Docker Compose, restart: unless-stopped, healthcheck per service
Watchdog	heartbeat table (60 s cadence); missing-heartbeat alert to Telegram; node-level checkpoint resume
Cloud state	Supabase free tier: Postgres 500 MB, Realtime, Storage, Edge cron (EOD fallback snapshot)
Secrets	.env on laptop (never committed); Supabase Vault for cloud-side keys; exchange API keys read-only scope
Backup	Supabase daily logical backup; weekly pg_dump downloaded to laptop; skills + profiles in git
Laptop-off behavior	Cloud EOD snapshot keeps brief archive warm; on reconnect the worker catches up from vendor APIs and replays missed triggers
Firebase alternative	Firestore adapter + Cloud Functions for EOD job (540 s limit shapes it to snapshot-only)

Table 13 — Deployment configuration summary

10.2 Secrets and key hygiene

Secrets follow a least-privilege, machine-local doctrine. LLM API keys live only in the laptop .env and are injected into containers at runtime; the Supabase service key never appears on the dashboard (which uses the anon key plus row-level security bound to the owner's auth UID); exchange keys, when Phase 4 arrives, are created with trading-disabled-then-enabled scope progression and IP restrictions where the venue supports them. Hermes skill config declares required environment variables by name following the skill.yaml config convention observed in the MOSS factory skill, so a skill can fail loudly at install time when a key is missing instead of mysteriously at run time. No secret is ever logged: the trace redaction layer strips key-shaped strings from all bot transcripts before they reach the dashboard.

11. Non-Functional Requirements and Cost Budget

11.1 Performance and reliability

Performance targets are set at the shape of one user with a 20-ticker IDX watchlist and a 15-coin crypto watchlist, on a mid-range laptop. The analysis window (06:15 to 07:00 WIB) must absorb the full deep pipeline with headroom; the intraday loop must evaluate all trigger conditions within one 5-minute bar interval; and the Telegram round-trip from command acknowledgment to substantive answer must feel conversational rather than batch. Reliability targets recognize the single-worker reality: the system degrades loudly and gracefully, and nothing in the design assumes the laptop is immortal.

Category	Requirement	Target
Latency	Deep pipeline per ticker (fast tier)	p50 <= 100 s, p95 <= 180 s
Latency	20-ticker batch inside morning window	completes by 06:55 WIB
Latency	Intraday trigger evaluation	<= 60 s from bar close
Latency	Telegram command ack / answer	<= 3 s / <= 60 s
Reliability	Brief on-time delivery (trading days)	>= 95% over 30 days
Reliability	Pipeline run success (resume incl.)	>= 98%
Reliability	Recovery from laptop sleep/reconnect	auto-resume <= 5 min, no duplicate plans
Security	Secret exposure in logs/transcripts	0 occurrences (redaction tested)
Security	Dashboard write surface	preferences only; RLS enforced
Maintainability	Vendor or model swap	config change only; no code edits

Table 14 — Non-functional requirements

11.2 Cost budget

Cost is a design constraint, not an afterthought, because the product must run indefinitely on a personal budget. The dominant variable is LLM choice, and the research pricing sweep shows the spread across a realistic 20-ticker daily cycle (roughly 150k input and 25k output tokens) is two orders of magnitude — which is why the provider-agnostic adapter with per-role model pinning is a core requirement rather than a nicety. At the recommended default (fast tier = GPT-4o-mini or Gemini 2.5 Flash-Lite for analysts and debates, deep tier reserved for the two judges), the expected spend is a few dollars a month; even an all-premium configuration stays under coffee money.

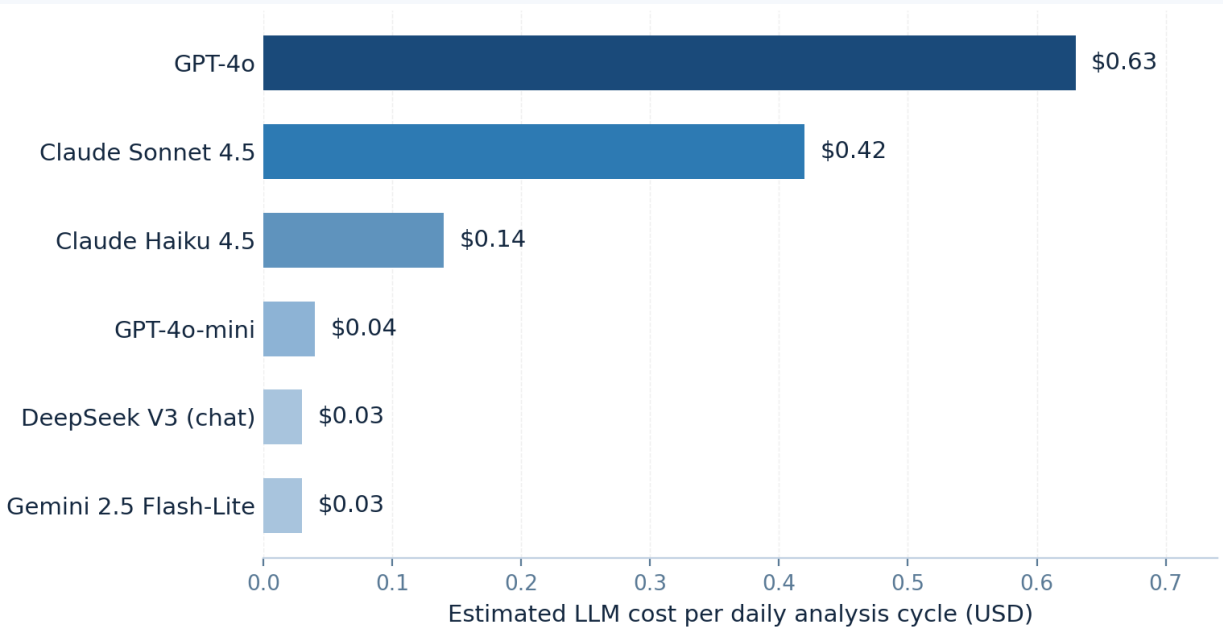


Figure 3 — Estimated LLM cost per daily analysis cycle by provider (20-ticker workload, Sep 2026 pricing)

Item	Assumption	Monthly cost
LLM API	20 tickers/day deep pipeline + debates + memory reflections	\$2-6
Supabase	Free tier (500 MB fits ~2 years of daily rows)	\$0
Vercel dashboard	Hobby tier	\$0
Telegram	Bot API free	\$0
Market data	yfinance + RSS + CoinGecko demo + ccxt public	\$0
Electricity / laptop	Always-on mini workload (~15 W average)	~\$1-2
Total	Recommended: fast tier + Sonnet judges	< \$10

Table 15 — Monthly operating budget (recommended configuration)

12. Roadmap and Milestones

Delivery is organized into five phases, each with hard exit criteria that gate the next. The sequence enforces the product's trust arc: data before analysis, analysis before advice, advice before measurement, measurement before automation. Dates are expressed in working weeks from project start, assuming one builder working evenings with AI-assisted development (the Hermes agent itself is expected to implement significant portions of each skill package from this document).

Phase	Weeks	Deliverables	Exit criteria
P0 Foundations	1-2	Compose stack up; Postgres schema migrated; yfinance/ccxt/ingest pollers; steward v0; heartbeat + kill-switch skeleton; MBG Lite Hermes bundle (F-11) published	10-year .JK history for 30 tickers cached with zero gaps; heartbeat visible in Supabase
P1 Daily brief MVP	3-4	Hermes profiles + mbg-brief skill; 4 analysts (no debate); reporter; /brief, /analyze, /status	Brief delivered 5 consecutive trading days by 07:05 WIB with all sections present
P2 Advice core	5-8	Full 13-node pipeline with debates; plan cards; Risk Guard rulebook; paper broker with T+2; vectorbt replay	20 plans paper-tracked over 10 sessions with correct fills/costs; guard rejection path proven
P3 Memory + dashboard	9-10	Reflection memory loop; dashboard views with Realtime; backtest run cards; walk-forward suite; F-10 Opportunity Scanner; event-timing gate in Risk Guard	First lessons injected into PM prompts and visible in transcripts; dashboard shows live equity
P4 Hardening + hooks	11+	Cost meters and budgets; validation sentinels CI; OJK-venue crypto semi-auto design doc; multi-asset expansion spike	30-day reliability report $\geq$ targets; semi-auto design reviewed and approved by user

Table 16 — Phase plan

Two scheduling principles protect the roadmap. First, P2 is the longest phase because it contains the three features users will actually judge the product by — plan quality, guard correctness, and paper-fill realism — and shortcuts there poison every later metric. Second, automation work (P4) cannot begin until the paper portfolio has produced at least 60 scored plans across varied market conditions; the review gate is evidence, not calendar. If real-world life delays a phase, scope shrinks before quality does: the non-goals list of Chapter 1 is the release valve.

13. Success Metrics and Acceptance Criteria

13.1 Key performance indicators

KPI	Definition	Target
Brief punctuality	% trading days brief lands $\leq$ 07:05 WIB	$\geq$ 95%
Pipeline success	% runs completing with resume incl.	$\geq$ 98%
Plan quality (paper)	% scored plans with positive alpha vs ^JKSE at T+5	$\geq$ 45% (honest coin-flip floor with positive expectancy)
Guard integrity	plans published violating any rulebook rule	0
Data integrity	mornings with unresolved steward anomalies	$\leq$ 1 per month
Cost discipline	mean daily LLM spend	$\leq$ \$0.35
Learning signal	memory lessons cited in later PM decisions	$\geq$ 60% of decisions

Table 17 — KPI register (measured from P2 onward)

The plan-quality KPI deserves explanation because it is deliberately modest. A 45% hit rate on direction with disciplined 1.5+ risk-reward produces positive expectancy; demanding 60%+ from an LLM swarm within three months would be a target written to fail. The KPI that actually guards product integrity is guard integrity at exactly zero — a single published plan that bypassed the risk rulebook is a critical defect regardless of whether it made money, because trust in the guardrails is the precondition for every future phase including automation.

13.2 Acceptance scenarios

Three end-to-end scenarios serve as the acceptance suite for the core features; each is executed verbatim during phase exit reviews and recorded with screenshots and transcripts.

- **AS-1 Morning brief** — Given the worker online and 20 tickers configured, when the 07:00 WIB cron fires on a trading day, then Telegram receives the full brief by 07:05 with every section present or explicitly degraded, the HTML archive URL opens, and the brief matches the analyses table contents (spot-checked on 3 tickers).
- **AS-2 Plan lifecycle** — Given a BBCA analysis producing a Buy verdict, when the plan card publishes with entry 9,800 / stop 9,500 / target 10,400 and size 6 lots, then Risk Guard validation passes, "Simulate fill" books a paper order at next open with 0.25% fee + levy, the position appears on the dashboard within one Realtime tick, and the T+2 ledger blocks a same-day reuse of the proceeds on a later sell.
- **AS-3 Memory loop** — Given a scored plan whose verdict was wrong ($\alpha < 0$ at T+5), when the steward resolves the pending memory entry, then a 2-4 sentence lesson is generated, the next analysis of the same ticker cites it in past_context, and the dashboard transcript shows the citation verbatim.

14. Risks, Assumptions and References

14.1 Risk register

Risk	L	I	Mitigation
yfinance endpoint changes or throttling breaks IDX bars	M	H	Fallback chain to scraper/stooq/local cache; bar-count alarm; batched polling with backoff
idx.co.id Cloudflare rules change silently	H	M	curl_cffi impersonation; golden fixtures per quarter; anomaly flagged in brief, not swallowed
LLM hallucinated prices or facts reach plan cards	M	H	Verified-snapshot grounding tool; structured outputs; guard rejects plans with ungrounded levels
LLM vendor price/model drift	M	L	Provider-agnostic adapter; per-role pins; cost meter with daily budget alarm
Overfitting daily loop to recent regime	M	M	Walk-forward gate before config changes; benchmark mandatory; memory lessons marked with regime context
Regulatory shift (OJK rules, IDX band changes)	M	M	Rules as config data; quarterly rules review task on roadmap
Laptop downtime misses sessions	H	L	Cloud EOD fallback snapshot; catch-up replay on reconnect; missed sessions scored as no-trade
Single-builder bus factor	M	M	Everything in git; skills portable; this PRD is the recovery manual

Table 18 — Risk register

14.2 Assumptions

- Single-user deployment; one Telegram owner; no multi-tenancy ever required.
- User's laptop runs Docker and is powered during market hours most trading days; cloud fallback covers the rest.
- IDX retail broker APIs remain unavailable through v1; if one launches, execution adapts behind the paper broker interface.
- LLM prices stay within an order of magnitude of September 2026 published rates.
- yfinance .JK coverage continues at current quality; a paid IDX feed remains an optional upgrade, not a dependency.

14.3 Open questions

- Should the sentiment analyst consume Stockbit community streams via unofficial means, or rely on X/RSS proxies only? (ToS risk vs signal value — decision deferred to P3 review.)
- Which IDR stablecoin quote convention should crypto paper P&L; standardize on — USDT via indodax, or direct IDR pairs?
- Does the Phase 4 crypto semi-auto mode require a separate OJK-compliant venue agreement or operate within existing personal accounts?
- Is a Bahasa Indonesia brief variant needed alongside English, and should it be a reporter config rather than a separate skill?

14.4 References

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